

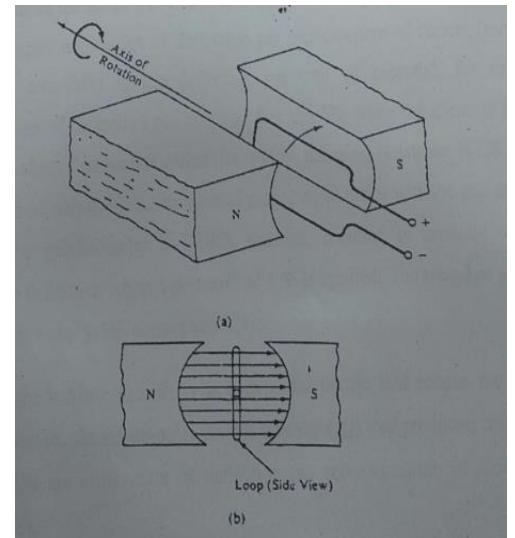
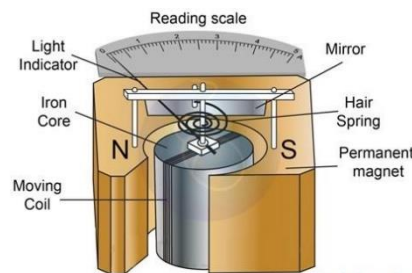
DC and AC deflection meter movement

Galvanometer: -

PMMC (Permanent Magnet Moving Coil)

Operation Principle:

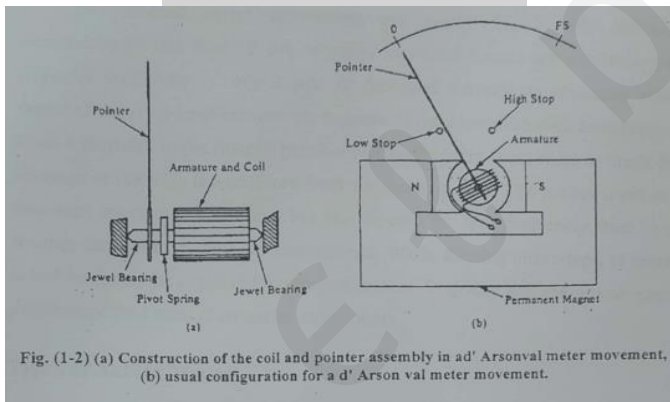
- A current in a wire generates a magnetic field with **polarity** determined by the current's direction and **strength** by its magnitude.
- **In a PMMC galvanometer**, the coil rotates between magnetic poles. The created magnetic field interacts with the poles' field.
- Current in one direction causes clockwise rotation; the opposite direction causes counterclockwise rotation.
- **The rotational position change** is proportional to the current magnitude.



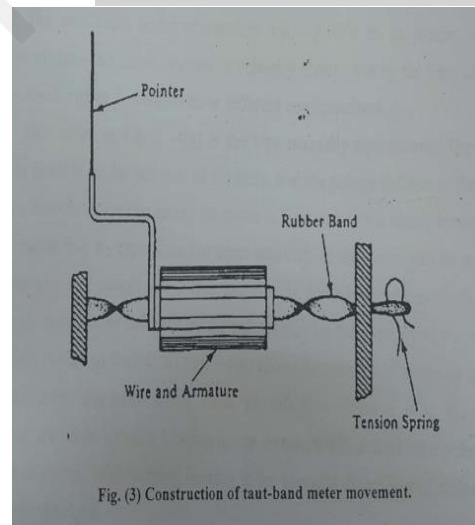
The sensitivity of a galvanometer can be defined in several different ways:

1. **Current sensitivity (SI):** The ratio of the deflection of the meter movement to the current. (the deflection per unit of current) **(mm/μA)**
2. **Voltage sensitivity (Sv):** The ratio of the deflection of the meter movement to the potential(voltage). (the deflection per unit of potential) **(mm/mV)**
3. **Megohm sensitivity (SR):** The number of megohm in series with the galvanometer required to produce one scale division deflection when a potential 1V is applied, when the galvanometric CDRX (Critical Damping Resistance) shunted.
4. **Ballistic sensitivity (SQ):** The dynamic metric that relates the maximum deflection to the single-pulse charge quantity Q that produces the deflection. **(mm/μC)**

D'Arsonval Meter Movement



Taut-Band Meter Movement



Coil Mounting

- ✓ **Coil** wound on armature or bobbin, mounted on jewel bearings to reduce friction.
- ✓ **Armature** suspended from fixed support on a stretched rubber band.

Armature Deflection

- ✓ **Current flow** in coil deflects armature proportional to current strength.
- ✓ **Band twists** as the armature rotates, no restoring force from pivot spring needed.

Pivot Spring

- ✓ **Coiled pivot spring** dampens pointer movement and returns it to zero when current ceases.
- ✓ **Taut-band movement** eliminates the need for a pivot spring.

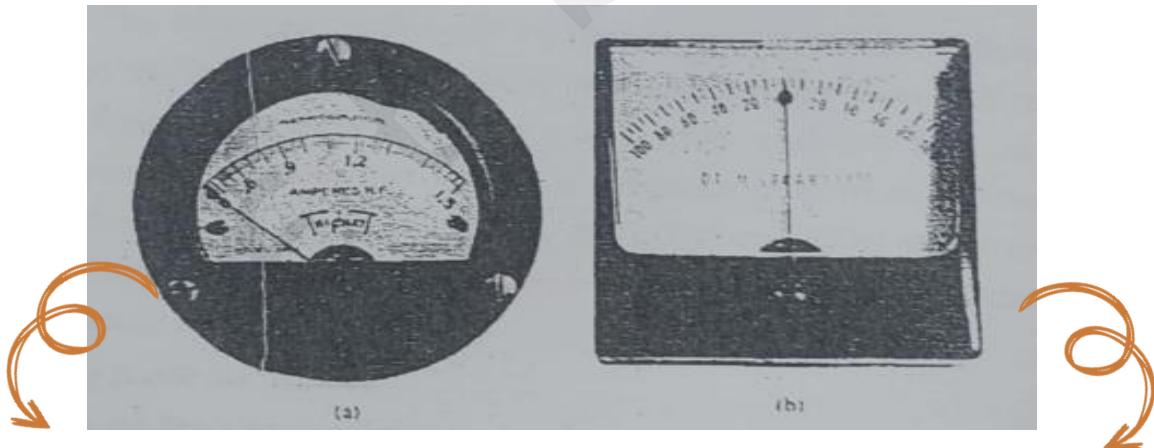
Advantages and Disadvantages

- ✓ **the taut-band** has two advantages over the older d'arsonval greater **sensitivity** and **more durability**.
- ✓ **D'Arsonval meter movements** are rarely found with full-scale sensitivities of less than **50 μA** , while **taut-band meters** are available with full-scale sensitivity of **only 2 μA** .
- ✓ **D'Arsonval movements** are more **easily damaged** due to their jewel-bearing construction, whereas **taut-band movements** are **more resilient**.

Analog Meters

Types of Analog Meters:

Meters are classified by the scale used.



Zero-current mark on the left,
pointer deflects right with
current flow.

Zero-current mark in the center,
pointer deflects left or right
depending on current direction.

- In some devices there is a **positive sign (+)** in side and **negative sign (-)** in the other side .(the positive sign is sufficient.)
- On **the scale edges** of the galvanometer there are **alarm sensors** that indicate when the pointer deflection is below a low limit or above a high limit.
- Other meter movements with **alarm capability** have an internal potentiometer element that outputs a resistance proportional to the pointer deflection.

Using Basic DC Meters: -

There are three basic rules for using the current meter:

1. Connect the meter in **series with the load or circuit** in which the **current** is being measured.
2. Use a meter with **a full-scale current rating that is greater** than the maximum current expected.
3. Use a meter that has **an internal resistance that is low compared** with the resistance in which it is being used **(less than 1:10)**.

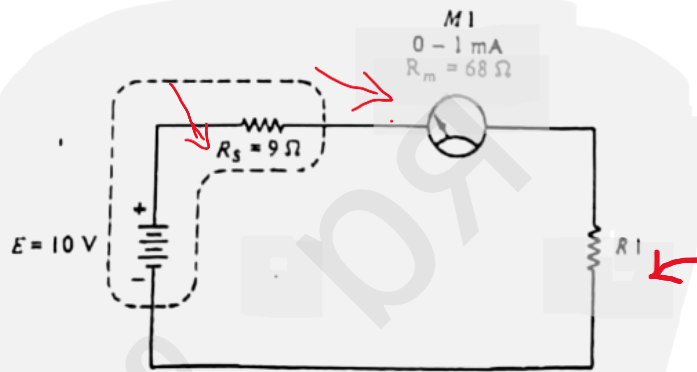


Fig. (1-5) Circuit with 0-to-1 -mA milliammeter.

The current flowing in meter $M1$ is given by

$$I_{M1} = \frac{E}{R_s + R_m + R_l} \quad (1-1)$$

where

I_{M1} = the meter current in amperes (A)

E = the open-circuit voltage in volts (V)

R_s = the power supply source resistance in ohms (Ω)

R_l = the load resistance in ohms (Ω).

In most cases, R_s and R_m are very small with respect to R_l and so are considered negligible. If this is true, then Equation (1-1) is reduced to

$$I_{M1} = \frac{E}{R_l} \quad (1-2)$$

Example : 1-2

Find the current flowing in meter $M1$ in Fig. 1-5 if $M1$ when $R_l = 15 \text{ k}\Omega$.

Solution . . .

Since R_l is very much greater than R_s or R_m , use Equation (1-2):

$$\begin{aligned} I_{M1} &= \frac{E}{R_l} \\ &= \frac{10 \text{ V}}{1.5 \times 10^4} \\ &= 6.7 \times 10^{-4} \text{ A} = 0.67 \text{ mA} \end{aligned}$$

Example : 1-2

Find the current that would flow in meter M_1 in Fig. 1-5 if M_1 were *incorrectly* connected in parallel with R_1 instead of in series as is proper.

Solution

Use Equation (1-1) because R_1 is very much greater than R_s and R_m , so its parallel effect is negligible. Delete R_1 from the equation:

$$I_{M1} = \frac{E}{R_s + R_m}$$
$$= \frac{10 \text{ V}}{(0 + 68) \Omega} = 0.13 \text{ A} = 130 \text{ mA}$$

Connecting M_1 across the load instead of in series results in a current that is 130 times too high for the full-scale rating. This current will probably destroy the instrument.

If we reduce **R1 to 150 Ω** and reduce **E to 100mV ... ?!!**

Without meter $I = E/R_1 = 0.1/150 = 0.00067 \text{ A} = \underline{0.67 \text{ mA}}$

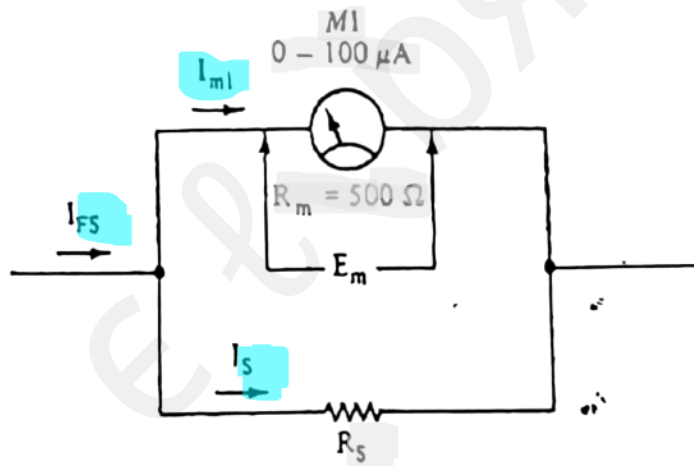
With meter $I = E/(R_1 + R_m) = 10/(150 + 68) = 0.00046 = \underline{0.46 \text{ mA}}$

>>>> big difference ~~XXXX~~

31 % reading error

Obtaining Higher Current Scales

By Using the shunt resistor.



$$I_{m1} = I_{FS} \left[\frac{R_s}{R_m + R_s} \right]$$

Example using the shunt resistor is given in Fig. 3-6. The total full-scale current of the meter with shunt in place is given by

$$I_{FS} = I_{M1} + I_S \quad (1-3)$$

where I_{FS} = the total current

I_{M1} = the current flowing in the meter

I_s = the current flowing in the shunt

Example (1-3) : The circuit of Fig. 1-6 is used to measure 1 mA full scale. How much current at full scale flows in (a) $M2$ and (b) R_s ?

Solution

Solve equation (1-3) for I_s :

$$I_{FS} = I_{M1} + I_S$$

Therefore :

$$\begin{aligned} I_S &= I_{FS} - I_{M1} \\ &= \left[1 \text{ mA} \times \frac{10^3 \mu\text{A}}{1 \text{ mA}} \right] - 10^2 \mu\text{A} \\ &= (10^3 - 10^2) \mu\text{A} = 900 \mu\text{A} \end{aligned}$$

There are two methods for calculating the value of resistor R_s if R_m , I_{M1} and I_{FS} are known: **Ohm's law** and the **current divider equation**.

Example 1-4

A 0-to 50 μA de meter movement, with a coil resistance R_m of 1250 Ω , is used with a shunt resistor to measure a full-scale current of 500 μA . Calculate the value of R_s .

$$I_s = I_{FS} \times \left[\frac{R_m}{R_m + R_s} \right]$$

$$(500 \mu\text{A} - 50 \mu\text{A}) = (500 \mu\text{A}) \times \left[\frac{1250 \Omega}{(1250 + R_s) \Omega} \right]$$

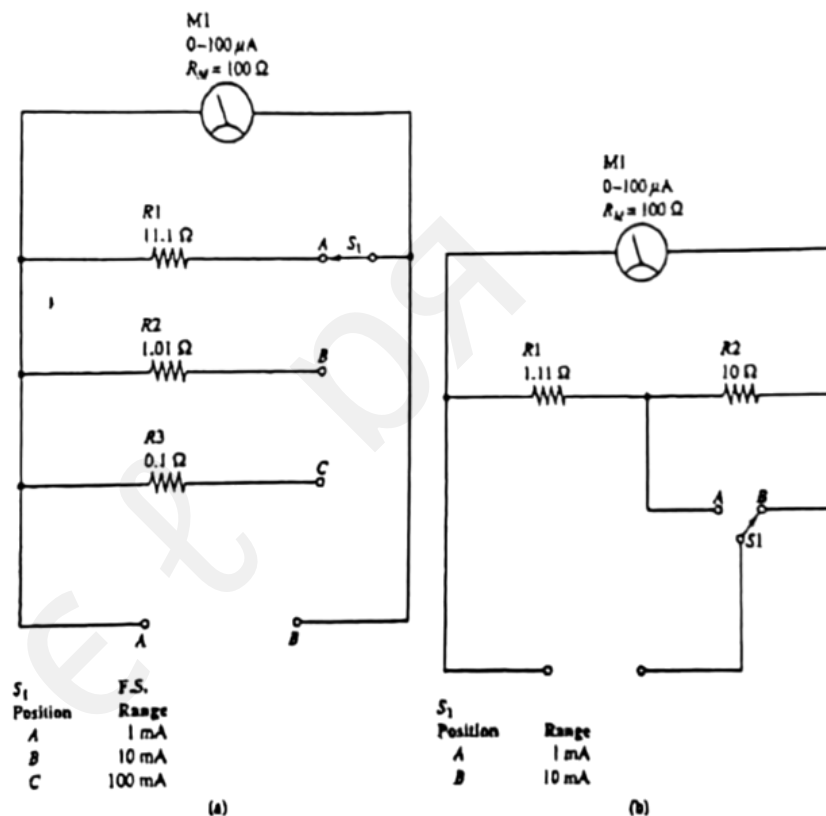
$$\frac{450 \mu\text{A}}{(500 \mu\text{A})(1250 \Omega)} = \frac{1}{(1250 + R_s) \Omega}$$

By algebraically rearranging the expression, we obtain

$$R_s = \left[\frac{(500)(1250)}{450} \Omega \right] - 1250 \Omega$$

$$= (1389 - 1250) \Omega = 139 \Omega$$

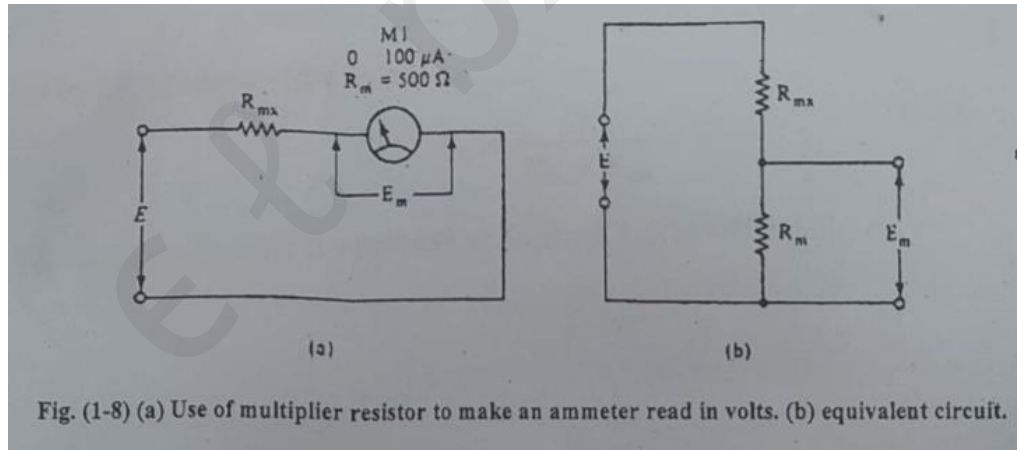
Although each method for determining the value of the shunt resistor has its own advocates, either method will yield the same answer. You are encouraged to work Example 1-4 using the Ohm's law method.



(a) Three-scale, multiple-range ammeter; (b) universal shunt, multiple-range ammeter

Voltage Measurements From DC Current Meters

Voltage can be measured on a DC current meter by connecting a multiplier resistor in series with the meter.



The formula to calculate current (I_1) is provided

$$I_1 = \frac{E}{R_m + R_{mx}}$$

where I_1 = the current in amperes (A)

E = the applied potential in volts (V)

R_m = the meter resistance in ohms (Ω)

R_{mx} = the multiplier resistance in ohms (Ω).

Example 5 :- Calculate the value of the multiplier resistor R_M in the figure (1-8)

-a for a full-scale potential E of 10V.

[- By ohm's law:

$$I = \frac{E}{R_m + R_{mx}}$$

$$10^{-4} \text{ A} = \frac{10 \text{ V}}{(500 + R_{mx}) \Omega}$$

$$10^{-5} \text{ A} = \frac{1 \text{ V}}{(500 + R_{mx}) \Omega}$$

By rearranging the expression, we obtain

$$R_{mx} + 500 \Omega = 10^5 \Omega$$

$$R_{mx} = 10^5 \Omega - 500 \Omega = 99,500 \Omega$$

[- By voltage divider equation:

$$E_m = \frac{ER_m}{R_m + R_{mx}}$$

$$0.050 \text{ V} = \frac{(10 \text{ V})(500 \Omega)}{(500 + R_{mx}) \Omega}$$

DC and AC Deflection Meter Movements With a little algebraic manipulation, we find

$$R_{mx} + 500 = \frac{(10)(500)}{0.050}$$

$$R_{mx} = (100,000 - 500) \Omega = 99,500 \Omega$$

The resistance is the same as before, as expected.

Voltmeter Sensitivity (ϕ) :- it is specified in terms of ohms per volt (Ω/V) .It is dependent upon the full-scale current range of the basic de meter movement used to make the voltmeter ,so :-

$$\phi = 1/I_{FS} \quad (\Omega/V)$$

DC Voltmeter Resistance (R_M) :- it is the product of its full-scale rating and the sensitivity rating ,so:-

$$R_M = E_{FS} * \phi \quad (\Omega)$$

Table (1-1) Sensitivity ratings for de voltmeters

Full-Scale Meter Current	Sensitivity (ϕ)
1 mA	1000 Ω/V
100 μA	10 k Ω/V
50 μA	20 k Ω/V
20 μA	50 k Ω/V
10 μA	100 k Ω/V

Example (1-6) :

Calculate the input resistance of a de voltmeter such as in Fig. 1-8(a) on the 0-500V de scale if the sensitivity is 10,000 Ω/V .

Solution

$$\begin{aligned} R_m &= E_{FS} \times \phi \\ &= (500 \text{ V})(10^4 \Omega/V) = 5 \times 10^6 \Omega = 5 \text{ M}\Omega \end{aligned}$$

Using Voltmeters :-

There are three basic rules for using voltmeters:-

1. Connect the **voltmeter in parallel** with the load.
2. Select a voltmeter that has **full-scale range that is greater** than the highest potential expected.
3. Make sure that the voltmeter has **an input resistance that is very high** compared with the circuit resistance (greater than 100:1).

Example 7 :- from this figure show if this voltmeter can use with this load or not.

:-

- $R_M = 5 \times 100K = 500 K\Omega$
- $R_{load} = 1M\Omega$

The ratio is 1:2

Without voltmeter :-

$$\begin{aligned} E_2 &= \frac{ER_2}{R_1 + R_2} \\ &= \frac{(10 \text{ V})(1 \text{ M}\Omega)}{1 \text{ M}\Omega + 0.5 \text{ M}\Omega} \\ &= \frac{10 \text{ V}}{1.5} = 6.67 \text{ V} \end{aligned}$$

With voltmeter :-

$$\begin{aligned} R_{eq} &= \frac{(1 \text{ M}\Omega)(0.5 \text{ M}\Omega)}{1 \text{ M}\Omega + 0.5 \text{ M}\Omega} = 0.33 \text{ M}\Omega \\ E_2 &= \frac{ER_{eq}}{R_1 + R_{eq}} \\ &= \frac{(10 \text{ V})(0.33 \text{ M}\Omega)}{(1.0 + 0.33) \text{ M}\Omega} = 2.48 \text{ V} \end{aligned}$$

big difference So this voltmeter isn't suitable.

